Dong Huu Cao

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Department of Computer Science
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EDUCATION

University of Illinois at Chicago, Chicago IL - *Doctor of Philosophy*

January 2023 - December 2027 (expected)

Major: Computer Science Major GPA: 4.0

Courses: Algorithms, Advanced Machine Learning, Graph Learning, Reinforcement Learning

Advisor: Prof. Sourav Medya

Earlham College, Richmond IN - *Bachelor of Arts*

August 2018 - November 2021

Major: Computer Science Major GPA: 4.0 Minors: Mathematics, Economics

Courses: Multivariate Calculus, Linear Algebra, Mathematical Analysis

Thesis: Unlabeled Consensus Modeler: Leveraging Voting Ensemble's Consensus on Unlabeled Data

Advisor: Prof. David Barbella

Honors: Phi Beta Kappa

RESEARCH/TEACHING EXPERIENCE

University of Illinois at Chicago, Chicago IL - *Research Assistant*

January 2023 - Present

- Research on graph learning, combinatorial optimization, and graph-based applications

University of Illinois at Chicago, Chicago IL - *Teaching Assistant*

August 2024 - Present

- Courses: Software Design (CS 342), Computer Algorithms I (CS 401)
- Grade exams and homework; hold office hours and lab sessions

Earlham College, Richmond IN - *Lead Student Researcher*

September 2020 - November 2021

- Goal of research: Collaborate with Green Filing, a local company, to categorize and extract information from legal documents
- Integrated several AWS services into an API wrapper for document parsing
- Arranged a pipeline to categorize documents of different types with 93.5% accuracy
- Set up an evaluation framework for the extraction of key items from the documents

Earlham College, Richmond IN - *Student Researcher*

May 2019 - July 2020

- Goal of research: Study the pricing of South Asian art
- Built Python bots that scraped auction sites to generate over 12,500 data records of art pieces

- Conducted a statistical analysis of the price distribution and its relations to other variables
- Implemented an interactive visualization dashboard of the data using Tableau
- Presented a research poster at Earlham EPIC Expo 2019

PUBLICATIONS

1. **Cao, D.**, Jannu, J., Ranu, S., Medya, S. TREEGEAR: Learning Graph Edit Distance with Zero Ground-truth Labels. (*in submission*)
2. **Cao, D.**, Velaj, Y., Medya, S.. LENS: Neighborhood-Preserving Partitioning for Exact Graph Unlearning. (*in submission*)
3. **Cao, D.**, Baghershahi, P., Chaturvedi, R., Medya, S.. A Graph-theoretic Approach to Privacy and Robustness in LLMs (*in submission*)
4. Shomee, H., **Cao, D.**, Medya, S.. Graph2Rule: Learning Behavioral Rules from Transformer Computation Graphs (*in submission*)
5. Tripathi, U., Dhaker, D., **Cao, D.**, Ranu, S., Medya, S.. Controlling LLMs via Reasoning Graph Counterfactuals (*ongoing work*)

INDUSTRIAL EXPERIENCE

Earlham College, Richmond IN - *Data Specialist*

January 2020 - May 2022 (full-time since December 2021)

- Manipulated data extracted from Oracle databases to produce 20+ interactive dashboards for the college's departments and offices
- Designed and incorporated a logical schema into the existing database structure for the tracking of utilities billing information

Sallie Mae, Indianapolis IN - *Architecture Intern (Xtern program)*

June 2021 - August 2021

- Built cloud apps that automated the manual workflows in Topic Modeling, PII Redaction, EDA, and Data Preprocessing using Amazon Comprehend and SageMaker Data Wrangler
- Co-developed the Application Search page as a proof of concept of the micro-frontends architecture on both the server side (C# ASP.NET) and client side (React)

PERSONAL PROJECTS

- Created an AI player of the board game Breakthrough, a chess variant (GitHub repo)
- Conducted independent research on the demographics of Garden Grove, CA (link to report)
- Developed a full-featured news application using Flask (GitHub repo)

SKILLS

Python (Numpy, PyTorch, Sklearn), SQL, AWS (S3, EC2, Comprehend, SageMaker, Textract), C, JavaScript, Linux, Git, LaTeX